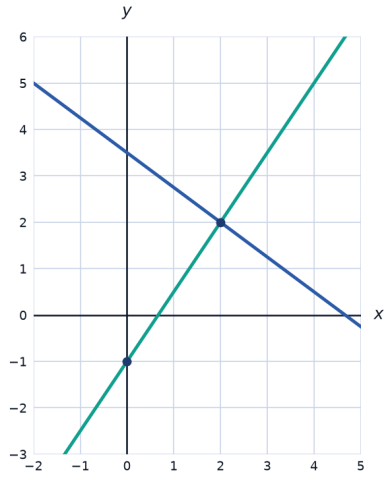
Algebra & Equations

Section 1: Quadratics & Complex Numbers

1 What is the value of the discriminant for the quadratic equation $2x^2 - 4x + 5 = 0$?



a. -24

b. 24

c. -56

d. 56

2 Simplify the complex number expression: i^{46}.

a. -1

b. 1

c. i

d. $-i$

3 What are the solutions of the equation $x^2 + 16 = 0$ over the complex numbers?

a. $\pm 4i$

b. ± 4

c. $\pm 2i$

d. $\pm 16i$

Section 2: Matrices & Determinants

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4 Find the determinant of the 2×2 matrix $A = [[4, 3], [2, 5]]$.

a. 14

b. 26

c. 10

d. 18

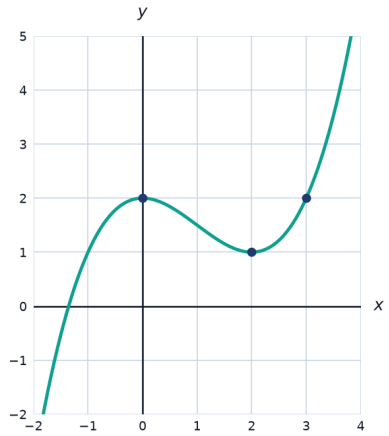
5 For what value of k does the matrix $M = [[k, 6], [2, 3]]$ have NO inverse?

a. 4

b. 6

c. 2

d. 3



6 If $A = [[1, 2], [3, 4]]$ and $B = [[2, 0], [1, 3]]$, what is the element in row 1, column 2 of $A + 2B$?

a. 2

b. 4

c. 6

d. 8

Section 3: Polynomials & Remainder Theorem

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7 What is the remainder when $P(x) = 2x^3 - 4x^2 + 5x - 3$ is divided by $(x - 2)$?

a. 7

b. 5

c. 9

d. 3

8 If $(x - 3)$ is a factor of $x^3 - kx^2 + 2x - 6$, what is the value of k?

a. 3

b. 2

c. 4

d. 5

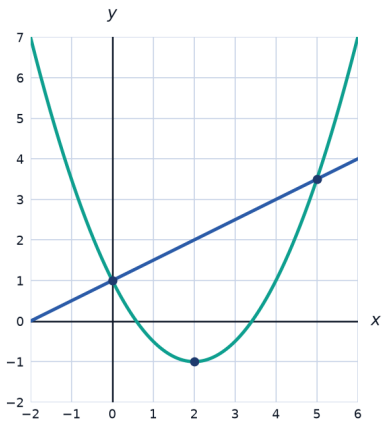
9 What is the degree of the polynomial $f(x) = (3x^2 - 1)^2(x^3 + 4)$?

a. 7

b. 6

c. 8

d. 12



Section 4: Systems of Equations & Inequalities

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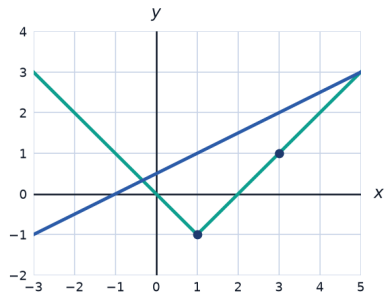
10 What is the value of $x + y$ for the system: $3x + 2y = 16$ and $2x - 2y = 4$?

a. 6

b. 5

c. 7

d. 8



11 Which point lies in the solution set of the system: $y > 2x - 1$ and $y \leq -x + 4$?

a. $(1, 2)$

b. $(2, 5)$

c. $(0, -2)$

d. $(3, 0)$

12 If $\frac{x-3}{x+2} = 0$, what is the real solution for x?

a. 3

b. -2

c. 0

d. 5

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